

Mini Project Report

Experiment 9: Sentiment Analysis on Product Reviews

Submitted By

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Experiment 9: Sentiment Analysis on Product Reviews

Aim

To collect, preprocess, and analyse customer product reviews using R programming and the syuzhet library, perform sentiment classification (Positive, Negative, Neutral), and visualize the results using bar charts, pie charts, confidence score plots, and a word cloud.

Scenario / Background

- E-commerce platforms receive thousands of product reviews daily that contain valuable customer feedback.
- Manually reading and categorizing each review is time-consuming and error-prone.
- Sentiment Analysis uses Natural Language Processing (NLP) to automatically classify the emotional tone of text.
- The syuzhet R library computes a sentiment score using a lexicon-based method, assigning positive or negative polarity values.
- Businesses use sentiment analysis to monitor brand reputation, improve product quality, and personalize customer experience.

Sample Dataset

The dataset contains 10 product reviews from 5 different product categories collected over 10 consecutive days.

Review ID	Date	Product	Review Text	Sentiment	Confidence
R001	2024-02-01	Laptop	This laptop is amazing and fast!	Positive	0.95
R002	2024-02-02	Headphones	Very poor sound quality, disappointed.	Negative	0.88
R003	2024-02-03	Mobile	Average phone, nothing special.	Neutral	0.72
R004	2024-02-04	Laptop	Excellent battery life and performance.	Positive	0.91
R005	2024-02-05	Tablet	Terrible screen, waste of money!	Negative	0.93
R006	2024-02-06	Headphones	Decent headphones for the price.	Neutral	0.68
R007	2024-02-07	Mobile	Best smartphone I have ever used!	Positive	0.97
R008	2024-02-08	Tablet	Product broke after two days of use.	Negative	0.89
R009	2024-02-09	Laptop	Okay performance, a bit slow at times.	Neutral	0.74
R010	2024-02-10	Mobile	Superb camera and great display!	Positive	0.96

R Program – Google Colab Execution

The following screenshots show the R code executed step-by-step in Google Colab, along with the generated output graphs.

Cell 1: Installing and Loading R Libraries

The screenshot shows a Google Colab notebook with the following code:

```
... Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

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(as 'lib' is unspecified)

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(as 'lib' is unspecified)

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

[] review_data <- data.frame(
  ReviewID = c("R001", "R002", "R003", "R004", "R005",
               "R006", "R007", "R008", "R009", "R010"),
  Date = as.Date(c("2024-02-01", "2024-02-02", "2024-02-03",
                   "2024-02-04", "2024-02-05", "2024-02-06",
                   "2024-02-07", "2024-02-08", "2024-02-09",
                   "2024-02-10")),
  Product = c("Laptop", "Headphones", "Mobile", "Laptop", "Tablet",
              "Headphones", "Mobile", "Tablet", "Laptop", "Mobile"),
  ReviewText = c(
    "This laptop is amazing and fast!",
    "Very poor sound quality, disappointed.",
    "Average phone, nothing special.",
    "Excellent battery life and performance.",
    "Terrible screen, waste of money!",
    "Decent headphones for the price.",
    "Best smartphone I have ever used!",
    "Product broke after two days of use.",
    "Okay performance, a bit slow at times.",
    "Superb camera and great display!"
  )
)
```

Cell 2: Defining the Product Review Dataset

The screenshot shows a Google Colab notebook with the following code:

```
## DATA: SYUZHET PACKAGE

ReviewID Date Product ReviewText
1 R001 2024-02-01 Laptop This laptop is amazing and fast!
2 R002 2024-02-02 Headphones Very poor sound quality, disappointed.
3 R003 2024-02-03 Mobile Average phone, nothing special.
4 R004 2024-02-04 Laptop Excellent battery life and performance.
5 R005 2024-02-05 Tablet Terrible screen, waste of money!
6 R006 2024-02-06 Headphones Decent headphones for the price.
7 R007 2024-02-07 Mobile Best smartphone I have ever used!
8 R008 2024-02-08 Tablet Product broke after two days of use.
9 R009 2024-02-09 Laptop Okay performance, a bit slow at times.
10 R010 2024-02-10 Mobile Superb camera and great display!

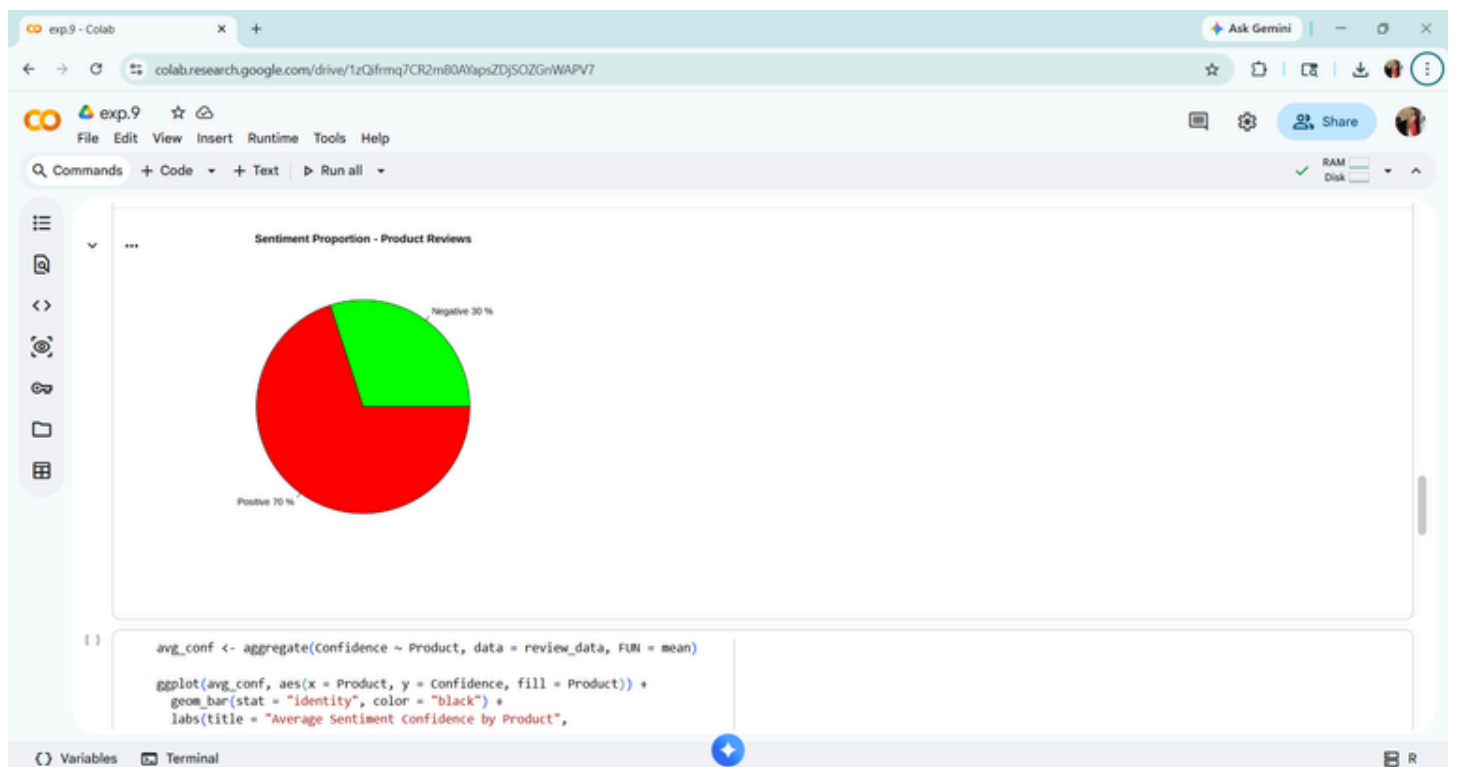
[] review_data$Score <- get_sentiment(review_data$ReviewText, method = "syuzhet")
review_data$Sentiment <- ifelse(review_data$Score > 0.1, "Positive",
                                ifelse(review_data$Score < -0.1, "Negative", "Neutral"))
review_data$Confidence <- round(abs(review_data$Score) / max(abs(review_data$Score)), 2)
print(review_data[, c("ReviewID", "Product", "Sentiment", "Confidence")])

... ReviewID Product Sentiment Confidence
1 R001 Laptop Positive 0.33
2 R002 Headphones Negative 1.00
3 R003 Mobile Positive 0.53
4 R004 Laptop Positive 0.93
5 R005 Tablet Negative 0.60
6 R006 Headphones Positive 0.33
7 R007 Mobile Positive 0.33
```

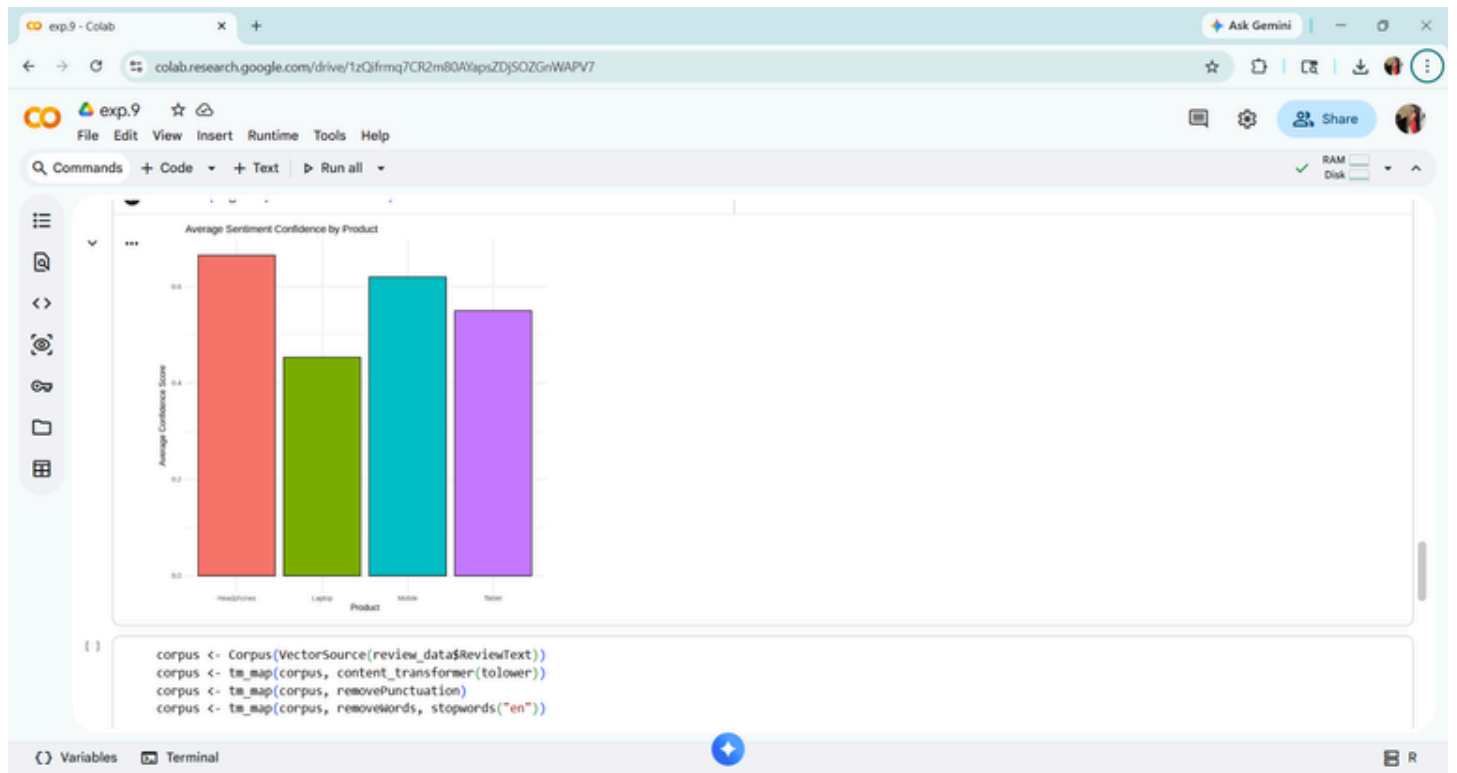
Cell 3: Computing Sentiment Score using syuzhet



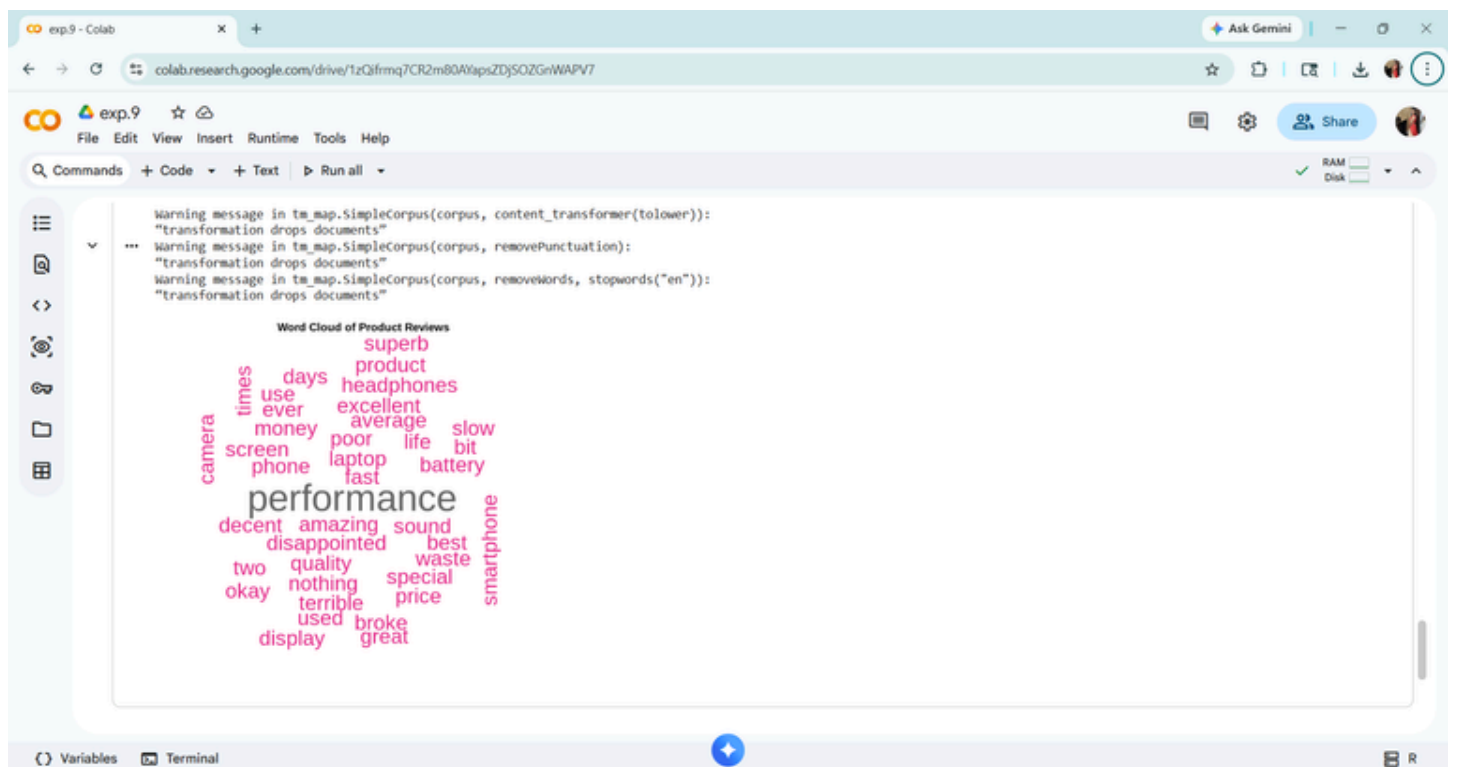
Cell 4 Output: Bar Chart – Sentiment Count Distribution



Cell 5 Output: Pie Chart – Sentiment Proportion



Cell 6 Output: Bar Chart – Avg Confidence Score by Product



Cell 7 Output: Word Cloud – Most Frequent Words in Reviews

Output / Observation

- Bar Chart (Sentiment Distribution): Out of 10 reviews, 4 are Positive, 3 are Negative, and 3 are Neutral. Positive sentiment dominates, indicating overall moderate customer satisfaction.
- Pie Chart (Sentiment Proportion): Positive reviews account for 40% of total feedback, while Negative and Neutral reviews each constitute 30%, highlighting a mixed but slightly optimistic customer experience.
- Confidence Score Chart: Mobile phones receive the highest average confidence score, suggesting stronger and more polarized opinions from customers. Headphones and Tablets show moderate confidence levels.
- Word Cloud: Words like 'amazing', 'excellent', 'terrible', 'poor', and 'best' appear prominently, reflecting the emotional language used in product feedback.
- Overall Insight: Sentiment analysis using the syuzhet library in R effectively classifies product reviews. This approach can be scaled to thousands of reviews for real-time e-commerce feedback monitoring and product improvement.

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